

ME 333 - MECHANICS OF MATERIALS
PRACTICE PROBLEMS 1 - CONCEPT OF STRAIN - AUG 16, 2025

Practice Problems:

From Prescribed Text - 2.54, 2.55, 2.57 (use eigen values), 2.58, 2.63 (use Mohr's circle), 2.68, 2.71, 2.72.

- (1) Use a computer to plot the undeformed and deformed state for a square plate with diagonally opposite corners at $(0, 0)$ and $(1, 1)$, given the displacement field:

(a) $u = 0.2 * \ln(1 + 3x + 2y), v = 0.4 * \exp(x)$

(b) $\mathbf{u} = (2x^3 + 0.5y^2) \hat{\mathbf{i}} + (x + 6y^2) \hat{\mathbf{j}}$

Note the following: a) both deformed and undeformed states are to be plotted on the same plot in different colors, b) consider a grid to present the deformed state of material points in the interior of the domain. Use a (10×10) grid.

- (2) A thin rectangular plate is rotated counter clockwise by 45° and then stretched along its length. The measured strain in the stretched direction is 0.625 (i.e., $5/8$), implying large deformation. If the position of point P after deformation is $(1, 1)$, find the coordinates of point P in the initial undeformed state ignoring Poisson effects.

[Answer] $\left(-\frac{1}{3\sqrt{2}}, \frac{5}{3\sqrt{2}}\right)$

- (3) Read the section "Rotation Between Two Line Elements (Definition of Shear Strain)" from the textbook and derive the relation:

$$\gamma_{12} = (1 + \epsilon_{E1})(1 + \epsilon_{E2}) \cos \theta^*$$

- (4) Show that the displacement field: $u = Ax + 3y, v = 3x - By, w = 5$, induces a state of plane strain ($\epsilon_{zz} = \gamma_{yz} = \gamma_{zx} = 0$). Also, determine the relationship between A and B for which the deformation is a constant volume deformation.

[Answer] $A = B$.

- (5) The deformation in a body is specified by $u = 4x - y + 3z, v = x + 7y$ and $w = -3x + 4y + 4z$. Determine the principal strains, principal directions and strain invariants.

[Answer] $\epsilon = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 7 & 2 \\ 0 & 2 & 4 \end{bmatrix}, \quad \epsilon_{(1,2,3)} = (3, 4, 8), \quad \mathbf{e}_{(1,2,3)} = \begin{bmatrix} 0 & 1 & 0 \\ 0.4472 & 0 & -0.8944 \\ -0.8944 & 0 & -0.4472 \end{bmatrix}, \quad \bar{I}_{(1,2,3)} = (15, 68, 96)$

- (6) The system of plane strain are given by the relations:

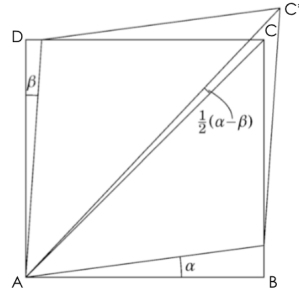
$$\epsilon_{xx} = A_0 + A_1(x^2 + y^2) + (x^4 + y^4); \quad \epsilon_{yy} = B_0 + B_1(x^2 + y^2) + (x^4 + y^4)$$

$$\gamma_{xy} = C_0 + C_1xy(x^2 + y^2 + C_2)$$

Determine the relation among constants for which the system of strains is permissible.

[Answer] $C_1 = 4, A_1 + B_1 = 2C_2 \quad \forall C_2$

- (7) Show that if the undeformed element as shown in the Figure is a square, the lower-left to upper-right diagonal rotates by an angle $\frac{1}{2}(\alpha - \beta)$, assuming that all angles are infinitesimal.



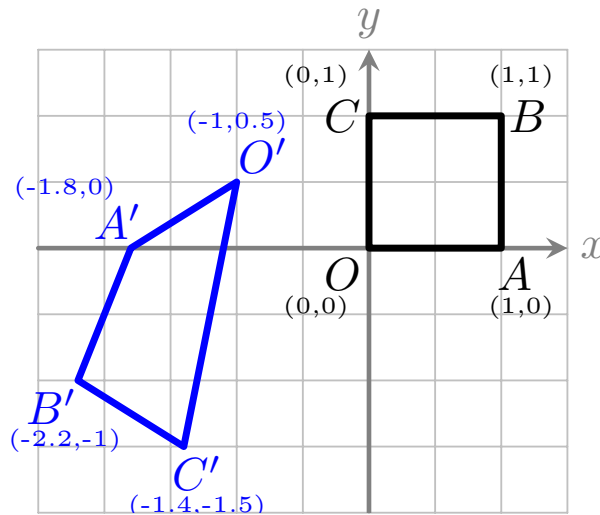
- (8) The following displacement field is imposed on a body:

$$\mathbf{u} = (yz\hat{\mathbf{i}} + 6y^2x\hat{\mathbf{j}} + 2zx\hat{\mathbf{k}})10^{-2}$$

Consider a point A and a neighboring point B where AB has the following direction cosines: $l_1 = 3/\sqrt{50}$, $m_1 = 4/\sqrt{50}$ and $n_1 = 5/\sqrt{50}$ with A located at $(2, 1, 3)$. If AB is Δs , find the components of A'B' and its direction cosines, post deformation.

[Answer] $(0.2\hat{\mathbf{i}} + 1.18\hat{\mathbf{j}} + 0.43\hat{\mathbf{k}})\frac{\Delta s}{\sqrt{50}}$, $(0.157, 0.928, 0.338)$

- (9) A square plate OABC (dashed - undeformed state) is subjected to a state of plane strain deforming it to O'A'B'C' as shown in figure below. Then: (a) Determine the displacement field (u, v) ; (b) Is the deformation homogeneous? Explain; (c) Determine the strain components using small strain theory. Are these strains accurate? - Explain; (d) Determine the strains using large strain theory.



[Answer] $u = -1 - 1.8x - 0.4y$, $v = 0.5 - 0.5x - 3y + xy$, No, normal strains are not physically realistic, $\epsilon_{xx} = 0.5y^2 - 0.5y - 0.055$, $\epsilon_{yy} = 0.5x^2 - 2x + 1.58$, $\epsilon_{xy} = -0.25x - y + 0.5xy + 0.66$